

## ASCII MASTER FLOW – PANEL 1/12: Ecosystem boundary card

BOUNDARY -> state the pond, forest patch, wetland or landscape being analysed

ABIOTIC -> climate, water, soil, minerals and topography inside that boundary

BIOTIC -> producers, consumers, detritivores and microbial decomposers

FUNCTION -> energy transfer, matter cycling, productivity and recovery

RULE -> a boundary is analytical; it must match the question and evidence

MUST REMEMBER: An ecosystem joins biotic communities and the abiotic environment through...

## ASCII MASTER FLOW – PANEL 2/12: Energy and matter split

SOLAR OR CHEMICAL INPUT -> producer fixation -> consumer transfer  
AT EACH TRANSFER -> respiration and heat dissipation reduce usable energy  
DEAD ORGANIC MATTER -> fragmentation -> microbial decomposition  
RELEASED NUTRIENTS -> abiotic pool -> producer uptake  
VERDICT -> energy moves one way; matter can cycle

## ASCII MASTER FLOW – PANEL 3/12: Functional guild ladder

PRODUCERS -> create organic matter from inorganic inputs  
PRIMARY CONSUMERS -> feed directly on producers  
HIGHER CONSUMERS -> feed through one or more trophic routes  
DETRITIVORES -> fragment dead material  
DECOMPOSERS -> chemically release nutrients

## ASCII MASTER FLOW – PANEL 4/12: Chain and web comparison

FOOD CHAIN -> one selected linear route

FOOD WEB -> connected set of chains within the same boundary

ALTERNATIVE LINKS -> may reduce dependence on one route

OMNIVORY -> one organism may feed across trophic positions

LIMIT -> connectivity does not immunise a system against every disturbance

## ASCII MASTER FLOW – PANEL 5/12: Habitat and niche matrix

HABITAT -> physical address

HABITAT NICHE -> where within that address the organism operates

TROPHIC NICHE -> resources and feeding relations

REPRODUCTIVE NICHE -> breeding requirements and timing

RULE -> overlap in place does not prove identical functional niches

## ASCII MASTER FLOW – PANEL 6/12: Ecotone logic

ECOSYSTEM A -> transition zone of variable width -> ECOSYSTEM B

OVERLAPPING CONDITIONS -> species from both sides may occur

NEW EDGE CONDITIONS -> some additional specialists may occur

COMPETITION AND TENSION -> the zone is not automatically benign

EDGE EFFECT -> possible density or variety pattern, not the definition

CLOSE DISTINCTION: Habitat is not niche, food chain is not food web, standing crop...

## ASCII MASTER FLOW – PANEL 7/12: Forest stratification section

CANOPY -> high light, wind exposure and arboreal niches

UNDERSTOREY -> filtered light and humid microclimate

SHRUB AND HERB LAYERS -> near-ground resource partitioning

FOREST FLOOR -> litter, detritivores and decomposer activity

CAUTION -> more layers do not prove universal stability

## ASCII MASTER FLOW – PANEL 8/12: Productivity accounting

GROSS PRIMARY PRODUCTIVITY -> total producer fixation per unit time

MINUS PRODUCER RESPIRATION -> metabolic energy used by producers

NET PRIMARY PRODUCTIVITY -> growth and production available onward

STANDING CROP -> living stock at a stated instant

RULE -> a stock cannot substitute for a production rate

## ASCII MASTER FLOW – PANEL 9/12: Resistance and resilience

RESISTANCE -> limited departure from prior condition during a disturbance

RESILIENCE -> recovery after the disturbance

MONOCULTURE -> may look uniform while remaining shock-specific and fragile

DIVERSE WEB -> may provide response options but no universal guarantee

ANSWER -> name the shock, response variable and time frame

## ASCII MASTER FLOW – PANEL 10/12: Services and contributions

PROVISIONING -> material outputs such as food, fibre and water

REGULATING -> flood buffering, pollination and climate moderation

SUPPORTING -> soil formation, primary production and nutrient cycling

CULTURAL -> spiritual, aesthetic, recreational and knowledge values

NCP -> includes relational and indigenous or local-knowledge values

## ASCII MASTER FLOW – PANEL 11/12: Carrying-capacity answer spine

DEFINE -> sustained load without erosion of regenerative functions

MEASURE -> regenerative supply, assimilative capacity and connectivity

CONTEXT -> consumption, technology and governance alter the threshold

LIMIT -> no timeless fixed headcount follows from the concept

CONCLUDE -> plan within ecological limits and monitor the response

## ASCII MASTER FLOW – PANEL 12/12: Evidence and institution firewall

BASIC OWNER -> definitions, mechanisms and verified PYQ concepts

ROUTING LEDGER -> paper, year and neutral demand ownership

OFFICIAL PAPER OCR -> printed wording only, never an inferred answer key

FSI OR SURVEY BODY -> extent or inventory within its stated method

LIVE STUB -> recorded as a failure, never converted into a current claim

EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Fix system boundary, trophic level, stock...